

Module 3 Reflective Journal

ITAI 1371 – Machine Learning

Neural Crew

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This lab helped me understand machine learning in a more complete way than I did before. Going into Module 3, I had heard the terms supervised, unsupervised, and reinforcement learning, but I did not completely understand what made them different from each other. At first, I was mostly trying to remember the definitions. After working through the examples in the lab, they started to make more sense. The easiest way for me to understand supervised learning was to think of it as learning with an answer key because the data already has labels. Unsupervised learning is different because the computer has to look through the data and find patterns or groups without being given the answers. Reinforcement learning also became easier to understand when I thought about it as learning through trial and error and receiving rewards or penalties based on the decisions that are made.

Another thing I learned from this lab was that there are several steps involved in building a machine learning model. Before doing this assignment, I did not realize how much work happens before a model actually makes a prediction. Looking at the Wine dataset helped me see why you first need to understand the data and make sure it is usable. The dataset had 178 samples, 13 features, three different classes, and no missing values. Seeing this information made me realize that checking and understanding the data is an important part of the process. If the data has problems or if you do not understand what you are working with, it could affect everything that comes after it.

The part of the lab that helped me understand this the most was experimenting with different features in the Wine classification model. I selected malic acid, flavanoids, and hue as my three features. My model reached about 83.3% accuracy, while the original model had about 88.9% accuracy. At first, I thought getting a lower accuracy meant that I had done something wrong. After looking at the results, I realized that this was actually showing me something important about machine learning. The features you choose can have an effect on how well a model performs. Just because data is available does not mean that every combination of features will give you the same results. This made feature selection feel more important to me than it did when I was just reading about it.

I also learned more about Logistic Regression during this lab. Before this assignment, the name sounded complicated because I am still fairly new to machine learning models. Seeing it actually work with the Wine dataset made it less confusing. The model was able to use information from the wine data to predict the different wine classes. I am still learning how the

math behind these models works, but being able to run the code and see the accuracy made the idea easier to understand. It also made me interested in seeing what would happen if I tried other combinations of features or completely different models.

One of the more challenging parts for me was understanding how all the different steps connect together. There is a lot happening between loading a dataset and getting a final prediction. I had to slow down and look at the outputs instead of just running the cells and moving to the next one. I am also still getting comfortable with Google Colab and Python code. When I first see a larger block of code, it can look confusing, but I am starting to get better at looking at smaller parts of it and understanding what they are doing. Running the code and seeing the results right underneath it has helped me learn because I can connect the code to an actual result.

Overall, this lab gave me a better understanding of what a machine learning workflow actually looks like. I still have a lot to learn, especially when it comes to choosing the best features and understanding how different models work. However, I feel more comfortable with these concepts than I did before completing the lab. The biggest thing I took away from this assignment is that building a machine learning model is not just about running code. The data you use, the features you select, the type of learning you choose, and how you evaluate the results all matter. Going forward, I would like to learn more about how to improve a model's accuracy and how someone decides which model and features are best for a real-world problem.